Practice Quiz: Systems

Part A: Multiple Choice

1. What is the primary role of **Systems** in Active Inference? A) To maximize reward B) To minimize variational free energy C) To increase entropy D) To eliminate the Markov Blanket

2. In Robotics, Systems is best described as: A) A static property B) A dynamic process C) An external state D) A random variable

3. Which mathematical quantity is most central to Systems? A) The Lagrangian B) The expected free energy C) The surprisal D) The precision

4. How does Systems relate to the concept of the Generative Model? A) It is separate from the model B) It is a component of the model C) It destroys the model D) It is only relevant for the environment

5. A failure in Systems would likely result in: A) Perfect prediction B) Generalized surprise C) Immediate death of the agent D) Zero entropy

6. Which scale is most relevant for analyzing Systems in this course? A) Quantum B) Neural C) Social D) All of the above

7. Systems connects directly to which other component? A) The step before it B) The step after it C) Both A and B D) None of the above

Part B: Short Answer

1. Explain how **Systems** facilitates the minimization of prediction error.
2. Provide a concrete example of Systems failing in a Robotics scenario.
3. How would you model Systems using a POMDP (Partially Observable Markov Decision Process)?